

Exploring the Impact of Vibrotactile Feedback on Boom Lift Safety Training: A Study Design

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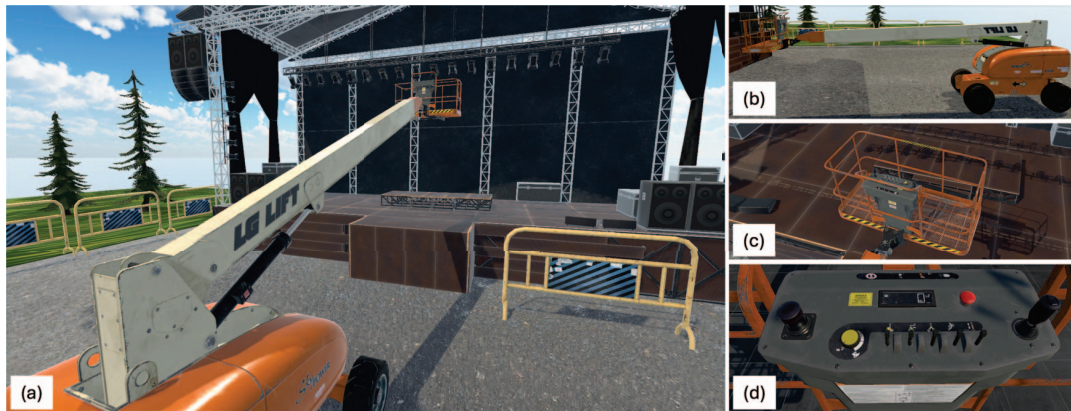
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Figure 1: The boom lift safe training simulation for our study. (a) performance stage, (b) boom lift, (c) platform, and (d) control panel.

ABSTRACT

In this paper, we present our research plan to investigate how various locations of vibrotactile feedback influence user experience and risk perception within a virtual reality (VR) based safety training simulation. We describe our planned user study where the study participants experience a construction boom lift ride (or operation) in an immersive VR environment. During the virtual lift ride, participants perform stage-maintenance tasks, such as assembling light bulbs and electrical lines, which involve certain safety risks—for example, hitting electrical lines and structures during the lift operation. In the study, the vibrotactile feedback condition is varied in three levels: (1) No-Vibration, (2) Controller-Vibration, and (3) Floor-Vibration. We plan to collect and analyze both subjective and objective data related to general user experience, sense of presence, and risk perception to examine the effectiveness of the safety training. Further details of our study and the boom lift ride simulation are described, and future plans are discussed.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies

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1 INTRODUCTION

Immersive technologies, such as virtual reality (VR), are increasingly being utilized in industrial and occupational training to create highly realistic practice environments [2, 6, 8]. With continuous advancements in multimodal VR technologies, including haptics, there is growing public and academic interest in these innovations. This underscores the necessity for rigorous and comprehensive research, focusing on user experience and human factors in the context of multimodal sensory feedback. Further exploration into the nuanced effects of these advancements on training effectiveness is crucial.

In this evolving landscape, one particular area requiring deeper exploration is the precise influence of haptic feedback on learning outcomes, skill retention, and overall training efficacy. Despite ongoing research efforts, the understanding of how haptic feedback specifically contributes to these aspects remains relatively underexplored. Our research aims to bridge this gap by focusing on the influence of *vibrotactile* feedback, a crucial haptic element, within the context of construction workers' safety training.

Previous studies were conducted to enhance the effectiveness of VR safety training. In particular, vibrotactile feedback was addressed as a key factor for increasing training effectiveness. For example, Moelmen et al. conducted a study on the educational effectiveness of multiplayer VR safety training at construction sites [5]. Interestingly, the results of this study revealed that participants with vibrotactile feedback positively impacted risk perception. However, their study did not consider the location of vibrotactile feedback, which can potentially impact the risk perception in safety training. Vibrotactile feedback can be provided at different locations on the body, such as through the controller [5], a wearable jacket [9], and the floor [3]. There is still a lack of understanding regarding the effects of vibrotactile feedback at different locations.

In this paper, we describe our plan to investigate the impact of

the provision and location of vibrotactile feedback on VR safety training effectiveness, focusing on aspects such as the sense of user experience, presence, and risk perception, through a comprehensive user study. For the study, we plan to develop a virtual boom lift ride simulation as a safety training scenario, where participants control the boom lift to perform stage maintenance tasks at height (Figure 1). The study involves three vibrotactile feedback conditions in a VR safety training simulation experience: (1) No-Vibration, (2) Controller-Vibration, and (3) Floor-Vibration. Regarding the effects of vibrotactile feedback on the safe training experience, our research focuses on the following research questions:

- RQ1.** Does the provision of vibrotactile feedback impact user experience and risk perception in safety training?
- RQ2.** Does the location of the vibrotactile feedback influence user experience and risk perception in safety training?

We describe the details of our study and the development of our boom lift riding simulation in VR, and discuss the future plans.

2 PLANNED STUDY

In this section, we provide an overview of our study and the details of settings, measures, and potential hypotheses.

2.1 Participants

We plan to recruit participants from the University of Calgary through online flyers and social media platforms for our study. To determine the appropriate sample size, a power analysis was conducted using G*Power, based on an ANOVA with repeated measures across three conditions. We anticipated a medium effect size, set the error probability (α) at 0.05, and aimed for a power of 0.80. The analysis suggested that a total sample size of 27 participants would be sufficient to detect the hypothesized effects with the desired level of statistical power.

2.2 Study Design

To investigate the effects of vibrotactile feedback on the safety training experience, specifically, the boom lift ride at height for stage maintenance tasks, we plan to conduct a within-subject study with three vibrotactile conditions: (1) No-Vibration, (2) Controller-Vibration, and (3) Floor-Vibration (Figure 2):

- **No-Vibration:** Participants will *not* experience any vibrotactile feedback during the boom lift ride experience.
- **Controller-Vibration:** Participants will experience vibrotactile feedback through the *hand-held VR controllers* during the boom lift ride experience.
- **Floor-Vibration:** Participants will experience vibrotactile feedback through the *floor platform* where they are standing during the boom lift ride experience.

In the conditions with vibrotactile feedback, the default vibration will be given, emulating the shaking of the boom lift platform. Additional vibrations will be provided upon collisions with risk factors. To ensure consistency across study conditions, the visual and auditory effects will remain the same, with only the vibrotactile feedback varying. To minimize potential learning effects from the task, the order of experiences will be counterbalanced among participants.

2.3 Stage Maintenance Tasks in VR

For the study, we plan to engage participants in a safety training simulation with stage maintenance tasks (Figure 1). The maintenance tasks in this study will involve a variety of activities, such as boom lift operation, risk factor avoidance, and stage equipment installation/maintenance.

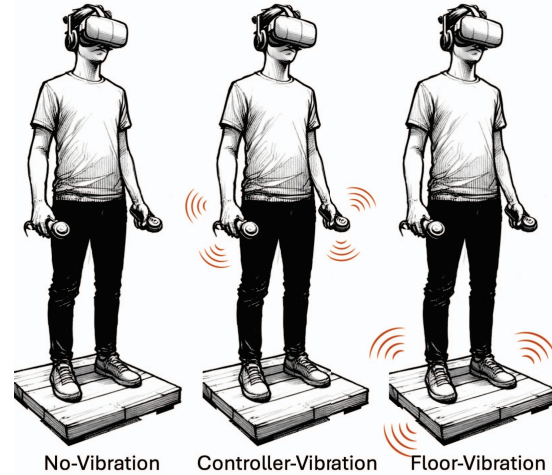


Figure 2: Vibrotactile feedback conditions.

Boom Lift Operation In a real-life scenario, operators use levers and control panel dials manually to operate a boom lift. Mimicking the real-life operation, participants will operate the virtual boom lift using hand-held VR controllers. Interacting with the control buttons on the virtual panel of the virtual boom lift, participants will be able to extend the boom lift platform and move it in different directions and orientations (Figure 1).

Risk Factor Avoidance While operating the boom lift in our virtual simulation, participants will be tasked with avoiding potential risks associated with maintenance tasks, such as colliding with electrical lines and navigating around obstacles on the ground or along their path. The simulation environment will replicate a partially constructed outdoor concert stage, complete with various electrical lines and structural elements. We will incorporate common risks associated with boom lift operation, such as electric shock, collisions with structures, and compression against ceilings, to provide a realistic training experience.

Stage Equipment Installation/Maintenance In the study scenario, participants will be tasked with using a boom lift to perform maintenance tasks, such as installing light bulbs and connecting cables on a stage. Specifically, they will operate the lift to reach various locations, where they must pick up stage light bulbs and connect electrical lines. The task requires them to grasp and maneuver the light bulbs and cables to predetermined spots while handling the lift's controls. To complete the task, participants will replicate these actions at three different, randomly assigned locations within the virtual environment. This approach aims to mitigate potential learning effects by ensuring variability in each trial.

2.4 Procedure

Participants will initially be briefed on the experiment by the researcher and asked to complete an informed consent form. After the consent, we will collect their demographic information and previous VR experience levels. Preparations will include a brief training session for 5–10 minutes, allowing participants to familiarize themselves with the VR headset and controllers, as well as the operation of the virtual boom lift. The training environment will be purposefully minimal, focusing exclusively on the lift operation without additional elements like a natural setting or performance stage, to prevent distractions and facilitate the participants' learning of lift operational skills. Following the training, participants will undertake the main tasks of the experiment, which involve stage maintenance activities, and will complete three boom lift operation sessions corresponding to the different study conditions. After each session, participants will remove the VR headset and respond to subjective

questionnaires to capture their experiences and perceptions. To mitigate fatigue and simulator sickness, there will be a five-minute break between sessions. Upon finishing all three conditions, we will conduct a brief interview to gather qualitative feedback on their experiences. The study will take approximately 60 minutes.

2.5 Measures and Hypotheses

In the study, we will collect and analyze both subjective and objective data to comprehensively investigate the effects of vibrotactile feedback in safety training. This includes participant questionnaire responses, task-relevant behaviors, and performance metrics.

Subjective Measures Among various measures, our study focuses on user experience, the sense of presence, and risk perception. The User Experience Questionnaire will be employed to collect the general user experience [4]. The Igroup Presence Questionnaire will be utilized for the subjective assessment of the presence [7]. Also, we will develop and use custom questions for the risk perception, e.g., the perceived height, based on previous literature [1].

Objective Measures To correlate the participants' task performance and behavioral data with the reported subjective perceptions, we will collect the task completion time, which is the total time taken to complete the tasks, as a measure of task performance. While logging the time stamps, the participants' motion activities will be recorded through the positions and rotations of the worn headset and the hand-held controllers. This will enable us to collect the number of instances where and when they encounter wires, structures, and ceiling collisions as errors during the tasks. As quantitative data, all collision events that occur while the user is testing in a virtual environment are logged, such as collision location, collided object, collision time, and user's viewpoint at the time of the collision.

Hypotheses Based on our expectation of the generally positive effects of the vibrotactile feedback and its appropriate location, we propose the following high-level hypotheses:

- H1.** Vibrotactile feedback will increase the level of user experience, presence, and risk perception (i.e., Controller-Vibration and Floor-Vibration > No-Vibration).
- H2.** Vibrotactile feedback at an appropriate location will induce a higher level of user experience, presence, and risk perception than the feedback at a different location of the body (i.e., Floor-Vibration > Controller-Vibration).

3 DEVELOPMENT OF VIRTUAL BOOM LIFT SIMULATION

For our study, we are currently developing a multimodal boom lift simulation in an immersive VR environment, which provides not only visual and audio stimuli, but vibrotactile feedback. This section describes the on-going development of our boom lift ride program.

The simulation program is developed using the cross-platform game development engine, Unity (v2022.3.15f1). We use a virtual boom lift model that is well-rigged with realistic animations purchased from the Unity Asset Store¹. This realistic boom lift model has various functionalities that users can practice the actual lift operation. For example, using the buttons on the virtual control panel (Figure 1), users can control the lift in the manner they operate a real lift—e.g., platform yaw, pitch, arm lift, arm yaw, and arm extension movements. To create a realistic stage maintenance environment, an outdoor concert stage under construction has been constructed using a virtual stage asset and other environment models². Figure 1

¹Animating Boom Lift, Kobra Game Studios: <https://assetstore.unity.com/packages/3d/vehicles/animating-boom-lift-aaa-quality-pbr-materials-124526> (Accessed 2024-01-13)

²Stage constructor, S48games <https://assetstore.unity.com/packages/3d/environments/urban/stage-creator-186215> (Accessed 2024-01-13)

depicts the overall appearance of our VR safety training simulation with the boom lift. The implementation of the simulation will be deployed on the commercial VR headset, Meta Quest 3.

Vibrotactile feedback is provided to the users either through the hand-held controllers or using a floor-attached vibrotactile unit³, which is anchored to a standing platform where the users will be located during the boom lift experience. Realistic vibrations, aligned with the boom lift movements and collision events, are implemented to enhance the training experience.

4 CONCLUSION

As public interest in immersive technologies and multimodal devices for realistic virtual experiences grows rapidly, our research aims to address the understudied area of haptic feedback in VR safety training. We particularly focus on understanding how the provision and location of vibrotactile feedback affect user experience and risk perception during VR-based boom lift operation training. This paper details our study plan and describes the current development of the boom lift operation simulator designed for this research. As we actively develop the simulation system, we are also preparing documentation for research ethics approval at our institute. Based on the information provided in this paper, we anticipate further discussions on the current study and future research directions concerning the impact of haptic feedback in VR to enhance the effectiveness of safety training, which we plan to explore during the workshop.

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REFERENCES

- [1] D. Huppert, E. Grill, and T. Brandt. A new questionnaire for estimating the severity of visual height intolerance and acrophobia by a metric interval scale. *Frontiers in Neurology*, 8, 2017.
- [2] S. Joshi et al. Implementing virtual reality technology for safety training in the precast/prestressed concrete industry. *Applied ergonomics*, 90:103286, 2021.
- [3] S. Jung et al. Floor-vibration VR: Mitigating Cybersickness Using Whole-body Tactile Stimuli in Highly Realistic Vehicle Driving Experiences. *IEEE Transactions on Visualization and Computer Graphics*, 27(5):2669–2680, 2021.
- [4] B. Laugwitz, T. Held, and M. Schrepp. Construction and Evaluation of a User Experience Questionnaire. In *HCI and Usability for Education and Work*, pages 63–76. Springer Berlin Heidelberg, 2008.
- [5] I. Moelmen et al. Asymmetrical Multiplayer Serious Game and Vibrotactile Haptic Feedback for Safety in Virtual Reality to Demonstrate Construction Worker Exposure to Overhead Crane Loads. In *Proceedings of the 38th International Symposium on Automation and Robotics in Construction*, pages 613–620, 2021.
- [6] R. Sacks, A. Perlman, and R. Barak. Construction safety training using immersive virtual reality. *Construction Management and Economics*, 31(9):1005–1017, 2013.
- [7] T. Schubert, F. Friedmann, and H. Regenbrecht. The Experience of Presence: Factor Analytic Insights. *Presence: Teleoper. Virtual Environ.*, 10(3):266–281, 2001.
- [8] H. Zhang et al. Fuzzy comprehensive evaluation of virtual reality mine safety training system. *Safety Science*, 120:341–351, 2019.
- [9] L. Zhu, Q. Cao, and Y. Cai. Development of augmented reality serious games with a vibrotactile feedback jacket. *Virtual Reality & Intelligent Hardware*, 2(5):454–470, 2020.

³ButtKicker Gamer PLUS: <https://thebuttkicker.com/products/butt kicker-gamer-plus> (Accessed 2024-01-13)